

#Question 1: Surface Blasting Design Calculator#

```
#Blasting Diameter (BD)
#Average In-Hole Density(p)
#Spacing To Burden Ratio (S/B)
#Spacing (S)
#Bench Height (BH)
#Explosive Mass (M)
#Sub-Drill (U)
#Stemming Length (SL)
#Burden (B)
#Technical Powder Factor (PF)
```

```
D = float(input("BD = "))
p = float(input("p = "))
a = float(input("a = "))
S = int(input("S = "))
BH = float(input("BH = "))
Rock_Type = str(input("Rock Type (Hard, Medium, Soft, Very Soft):"))
```

```
U = 1/4*D
print("U = ", U)
M = (p*(D**2))/1273
print("M = ", M)
```

#Stemming Length Calculation

```
SL = round(20*D, 2)
```

```
print("SL = ", SL)
```

#Technical Powder Factor

```
if Rock_Type == "Hard":
```

```
    Tech_PF = 0.7
```

```
elif Rock_Type == "Medium":
```

```
    Tech_PF = 0.45
```

```
elif Rock_Type == "Soft":
```

```
    Tech_PF = 0.3
```

```
elif Rock_Type == "Very Soft":
```

```
    Tech_PF = 0.15
```

```
else:
```

```
    Tech_PF = None
```

```
    print("Invalid Rock Type")
```

```
if Tech_PF is not None:
```

```
    print("Technical Powder Factor =", Tech_PF)
```

```
while True:
```

```
    if SL >= BH:
```

```
        print("Error! SL cannot be greater than BH")
```

```
        BH = float(input("BH = "))
```

```
else:  
    break
```

```
#Burden Calculation
```

```
import math
```

```
B = float(round(math.sqrt((M*(BH-SL))/(a*Tech_PF*BH)), 2))  
print("Burden = ", B)
```

```
#Powder Factor Calculation
```

```
Actual_PF = float(round((BH-SL+U)*M/(B*S*BH), 4))  
print("Actual Powder Factor = ", Actual_PF)
```

```
#Question 2: Fragmentation Calculator
```

```
# Average In-Hole Density (p)
```

```
# Bench Height (BH)
```

```
# Relative Effective Energy of Explosive (E)
```

```
# Stemming Length (SL)
```

```
# Average Density
```

```
p = float(input("p = "))
```

```
# Bench Height
```

```
BH = int(input("BH = "))
```

```
# Stemming Length
```

```
SL = round(20*D, 2)
```

```
# Rock factor (A)
```

```
# Relative Effective Energy of Explosive (E)
```

```
# Mass of Explosive per Hole (M)
```

```
Rock_Type = str(input("Rock_Type: "))
```

```
print("A =", A)
```

```
Ex = str(input("Explosive Type:"))
```

```
print("Ex =", Ex)
```

```
if Rock_Type == "Hard":
```

```
    A = 14
```

```
elif Rock_Type == "Medium":
```

```
    A = 11
```

```
elif Rock_Type == "Soft":
```

```
    A = 9
```

```
elif Rock_Type == "Very Soft":
```

```
    A = 6
```

```
else:  
    print("Invalid Rock Type")
```

```
if Ex == "ANFO":
```

```
    E = 100
```

```
elif Ex == "TNT":
```

```
    E = 115
```

```
else:
```

```
    print("Invalid Explosive Type")
```

```
# Mean fragmentation size (X_50)
```

```
X_50 = round(float(A*(Tech_PF)**(-0.8) * M**0.167 * (115/E)**0.633),2)
```

```
print("X_50 =", X_50)
```